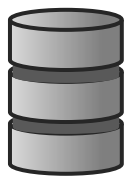
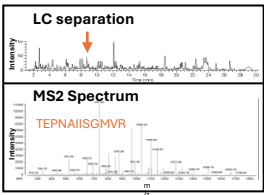


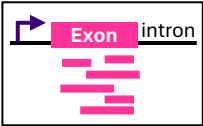
Public raw
LC-MS2
data



Proteome
database
search



GFF3-guided
peptide
projection



Apollo
JBrowse
deployment



Figure 1: Comparison of wheat proteogenomics workflows. (A) Previous workflow relying on public identifications followed by tBLASTn alignment against IWGSC RefSeq v2.1. [Vincent & Appels, 2024] (B) Genome-guided workflow developed in this study involving raw LC-MS/MS reanalysis against the IWGSC RefSeq v2.1 proteome, followed by GFF3-guided peptide projection and generation of browser-ready BED6/BED12 tracks for Apollo/JBrowse visualization; details are available in Suppl. Fig. S1 and Jupyter Notebook.

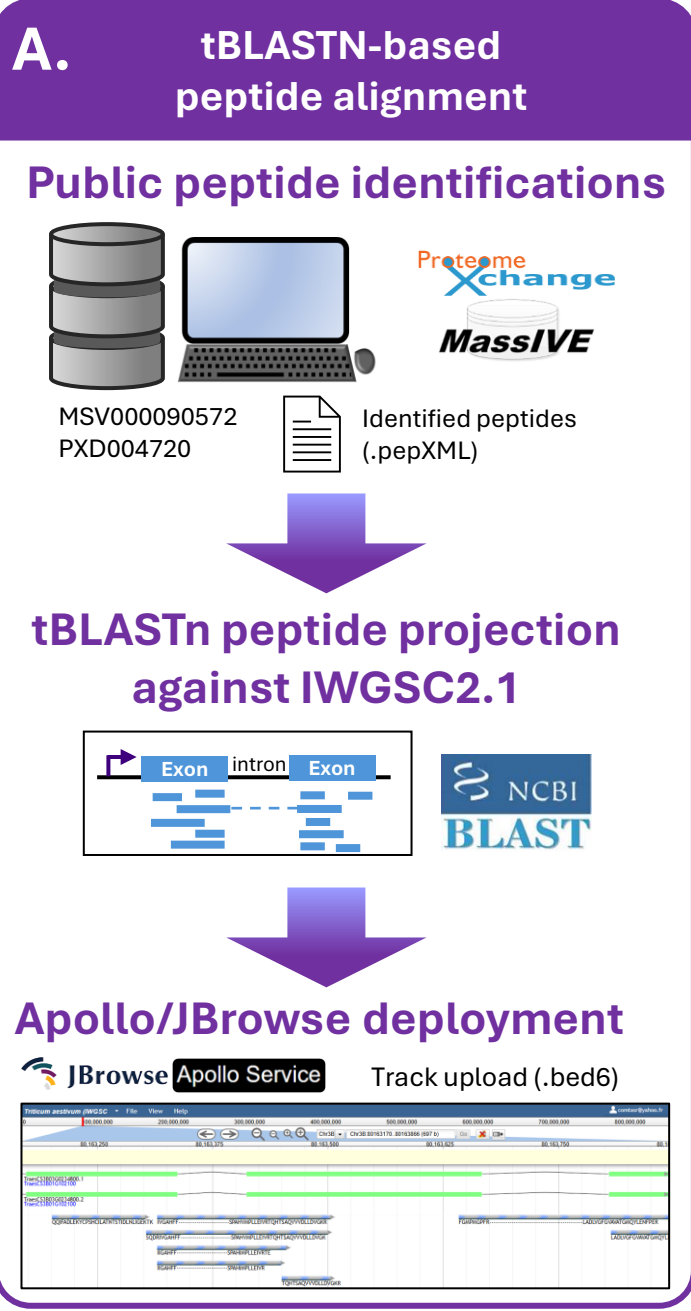
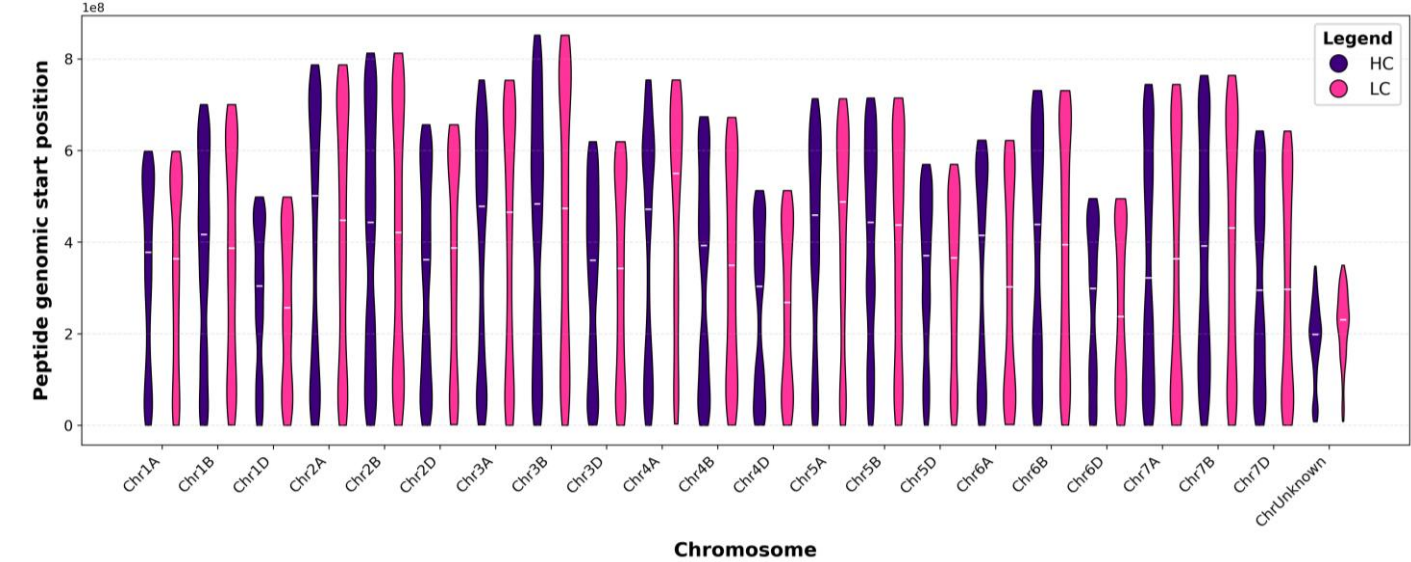


Figure 2: Genome-wide distribution of projected peptide evidence across wheat chromosomes. (A) Violin plot showing genomic coordinate distributions of HC and LC peptide projections for each chromosome. (B) Circular representation of tissue-specific projected peptide tracks across the wheat genome demonstrating extensive chromosomal coverage from multiple proteomics datasets.

A.



B.

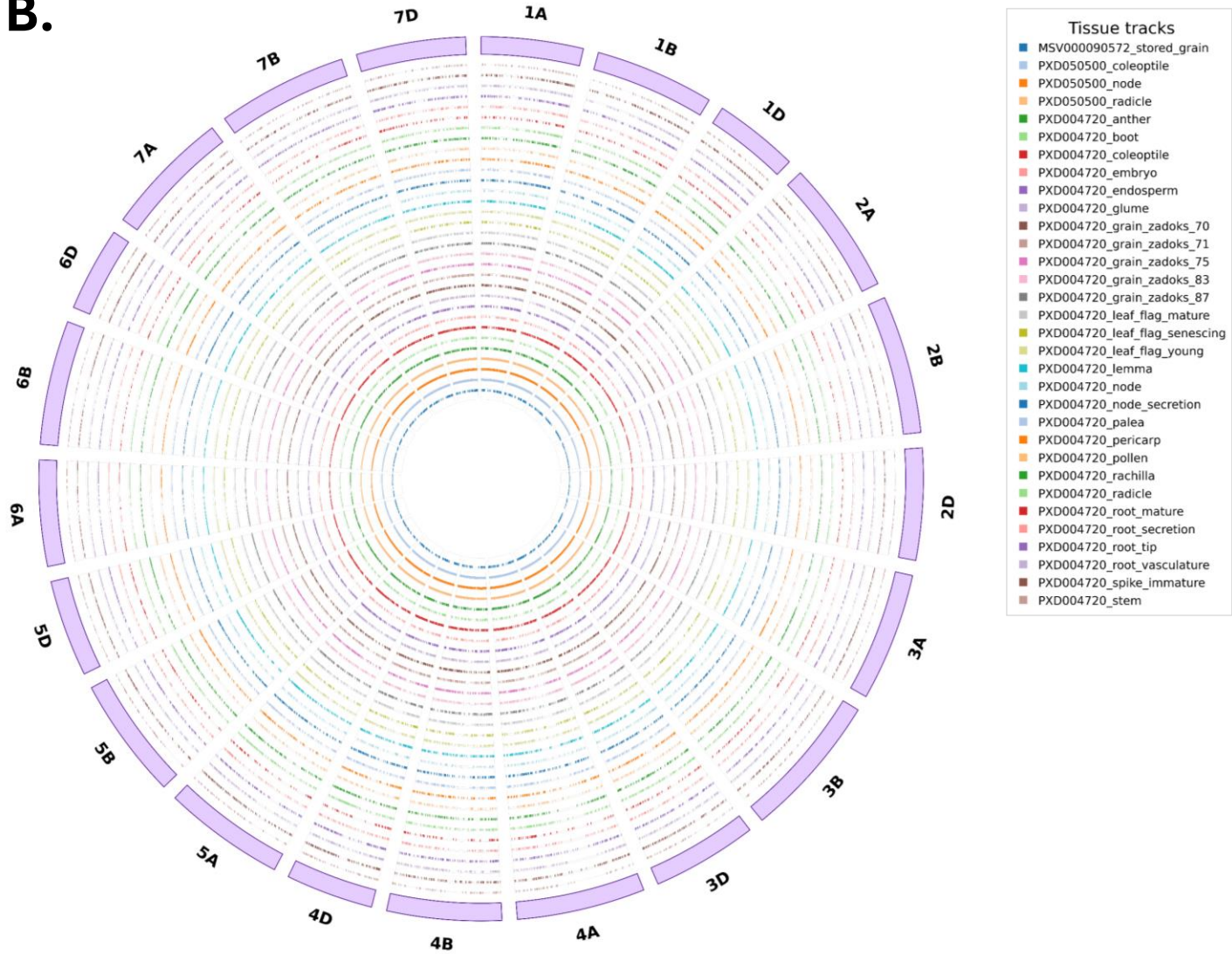
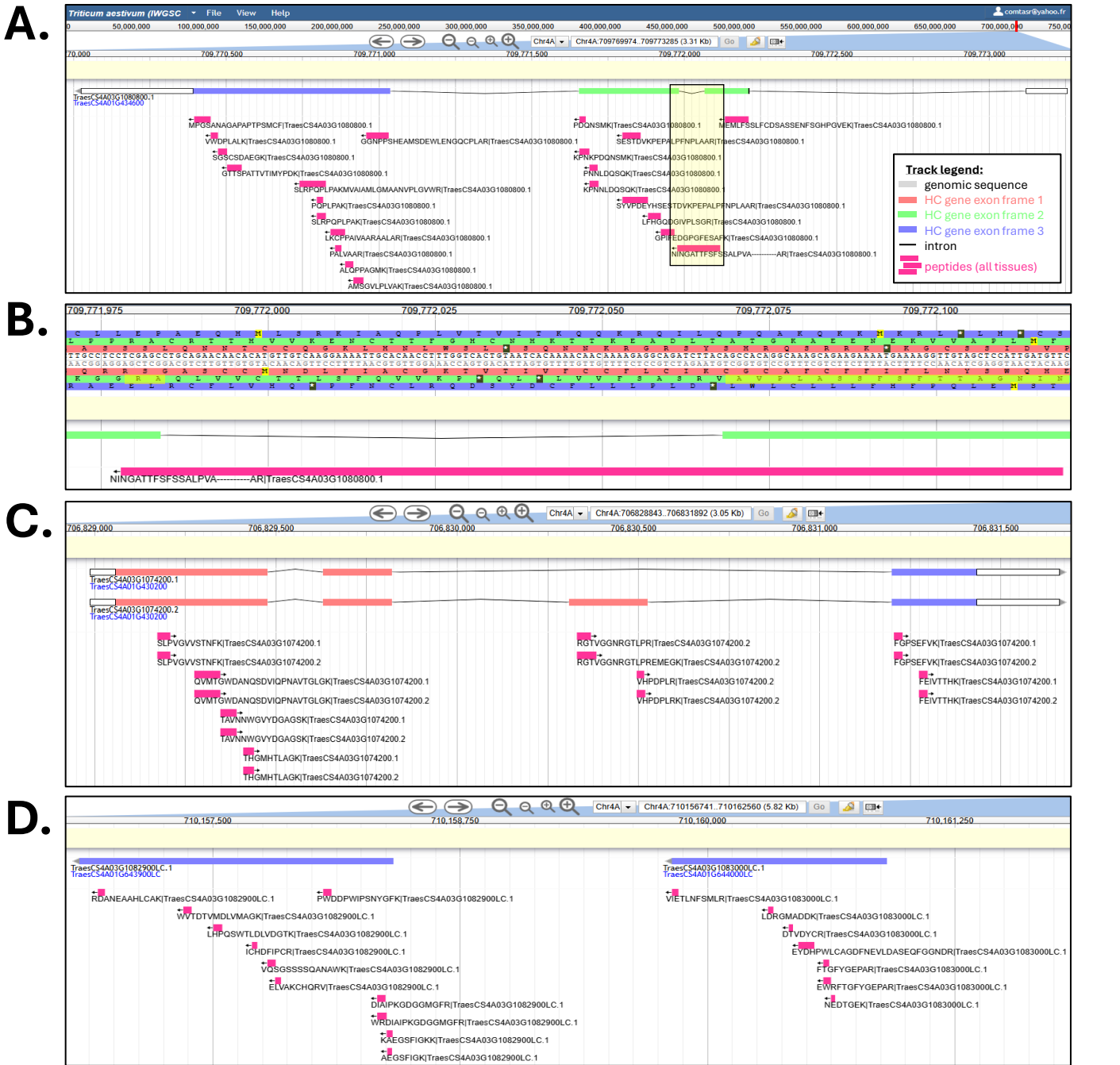


Figure 3: Apollo JBrowse visualization of annotation-aware peptide projections along wheat gene models. (A) Validation of HC gene model TraesCS4A03G1080800.1 supported by extensive projected peptide evidence(Chr4A:709769974..709773595). (B) Zoomed view of an exon-spanning peptide projection containing an intronic gap within TraesCS4A03G1080800.1 (Chr4A:709771972..709772129). (C) Peptide evidence supporting isoform-specific projections for TraesCS4A03G1074200.1 and TraesCS4A03G1074200.2 (Chr4A:706828843..706831892). (D) Peptide evidence supporting LC gene models TraesCS4A03G1082900LC.1 and TraesCS4A03G1083000LC.1 (Chr4A:710156743..710162562), suggesting potential promotion to HC status. Note: no peptide from Vincent & Appels aligned with the examples chosen in the figure.



Genome-guided wheat proteogenomics workflow

1. DATA RETRIEVAL + REFERENCE GENOME + PROTEOME SEARCH

Steps 1–4

FTP download of raw LC–MS2 proteomic and GFF3 genomic files (Unix)

Data conversion (ProteoWizard MSConvert)

Protein database creation with decoy (Galaxy)

Proteome search against IWGSC v2.1 database (FragPipe: MSFragger → PeptideProphet → ProteinProphet)

2. GFF3 ANNOTATION FRAMEWORK

Step 5

ProteinID → TranscriptID → GeneModel → CDS blocks (Python)

3. PEPTIDE EVIDENCE CONSTRUCTION

Steps 6–8

FragPipe output importation (Python)

Non-contaminant peptide–protein–gene evidence (Python)

Protein to gene model projections using GFF3-derived table (Python)

4. GFF3-GUIDED PEPTIDE PROJECTION

Step 9

Peptide in protein → coding nucleotide coordinates → exon-aware genomic blocks (Python)

Protein sequence
MKWVTFPEPTIDEISLLFSSAY...
↓
Peptide CDS coordinates
nt 241–312
↓
GFF3 exon blocks
Exon1 – Intron – Exon2
↓
Genomic peptide projection
[redacted]

1. Locates the peptide amino acid sequence within the corresponding protein sequence.
2. Converts the peptide amino acid interval into coding nucleotide positions.
3. Retrieves the CDS blocks associated with the corresponding transcript.
4. Projects coding nucleotide positions onto genomic coordinates.
5. Collapses projected nucleotide positions into genomic blocks.
6. Generates an intron-aware peptide display label (exon-spanning junctions shown using dash characters).

5. PEPTIDE PROJECTION VALIDATION

Steps 10, 25

Translation validation: projected genomic blocks → translated peptide → match/mismatch QC (Python)

Independent sanity checks (Python)

Projected genomic blocks
[redacted]
↓
Retrieved genomic DNA
ATGGCC...GT...AAGTAA
↓
Translated peptide
NINGATTFSSALPVA
↓
Original peptide
= NINGATTFSSALPVA
↓
Evaluation
☑ match

Beside the translation validation, 3 independent sanity checks are performed:

1. BED geometry check: confirms BED coordinates by checking start coordinates < end coordinates, and that BED block sizes and block starts are consistent with each peptide genomic interval.
2. Chromosome and strand check: confirms that projected peptides are assigned only to expected wheat chromosomes and valid strand values (‘+’ or ‘-’).
3. Protein-coordinate consistency check: confirms that each peptide projection has coherent amino-acid coordinates, including matching peptide length and valid protein coordinate intervals.

Evaluation:

- translated sequence match + successful checks → peptide validated ☑
- translated sequence mismatch and/or unsuccessful check → peptide rejected ☒

6. RESOURCE OUTPUT + INTERPRETATION

Steps 13–24, 26–28

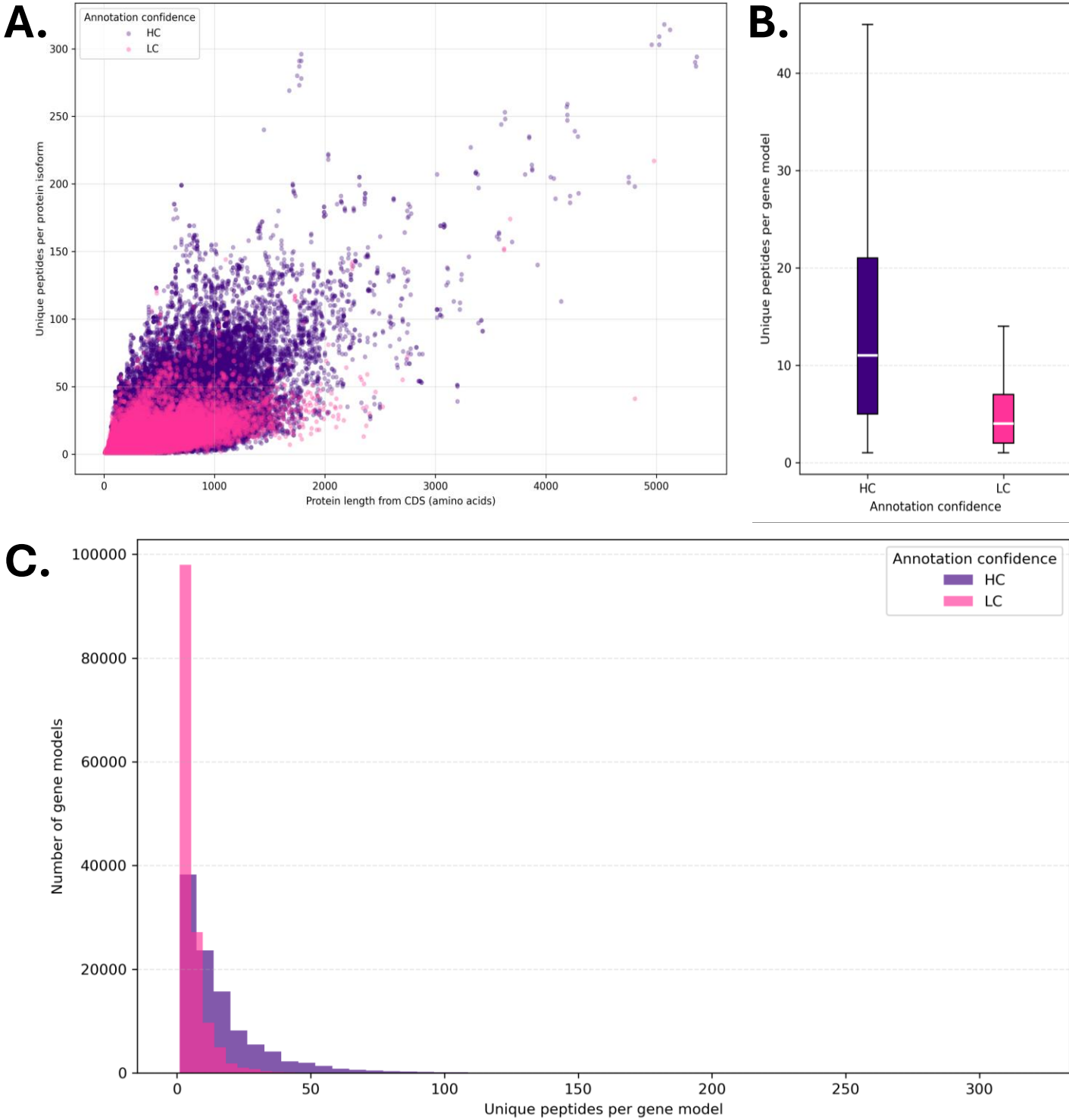
EDA (scatter, box, violin, bar, UpSet, circular plots, histograms, Venn diagram) (Python)

HC/LC support (Python)

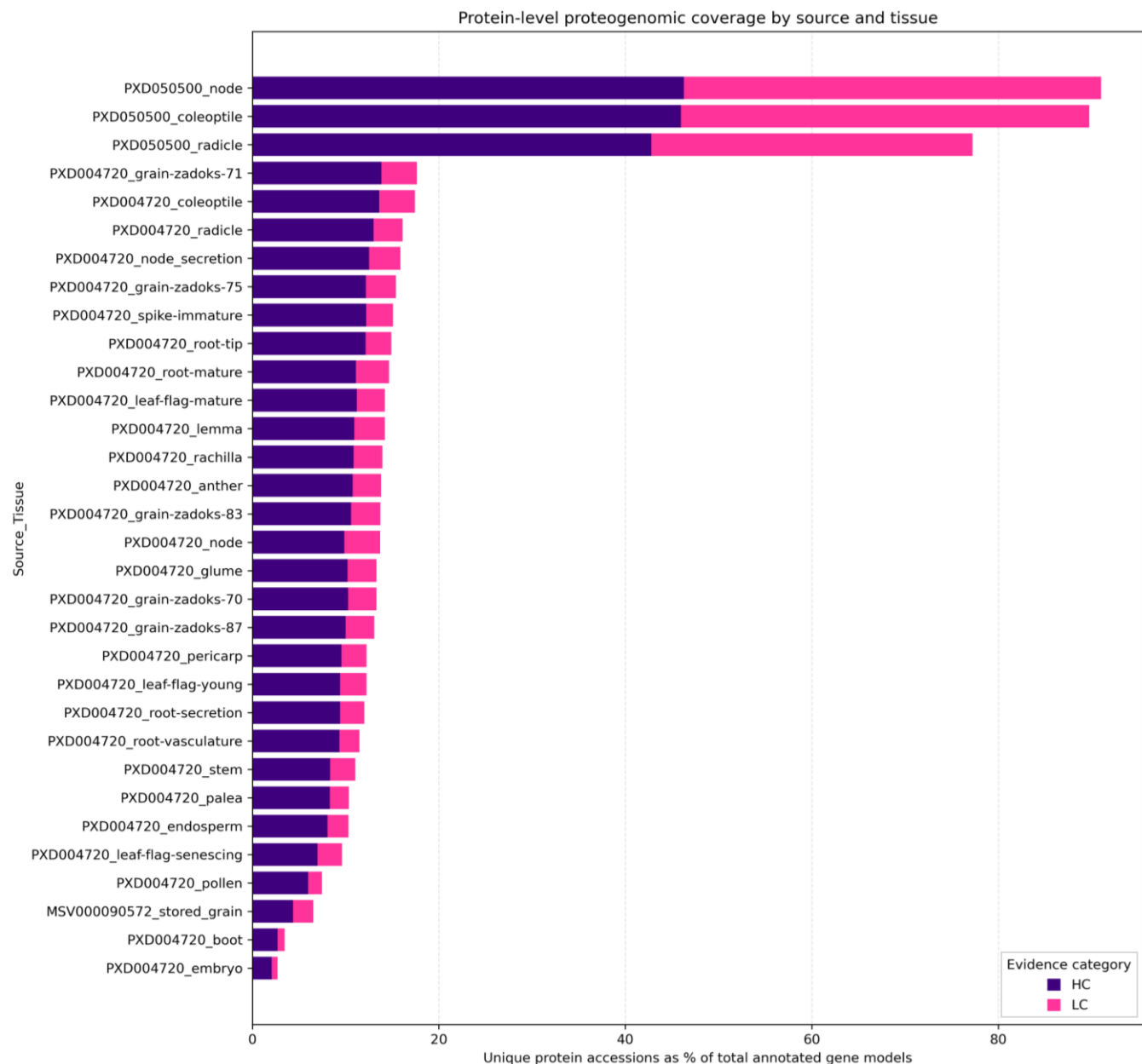
validated projections exported as BED tracks (Python)

BED tracks permanent upload (Apollo Jbrowse)

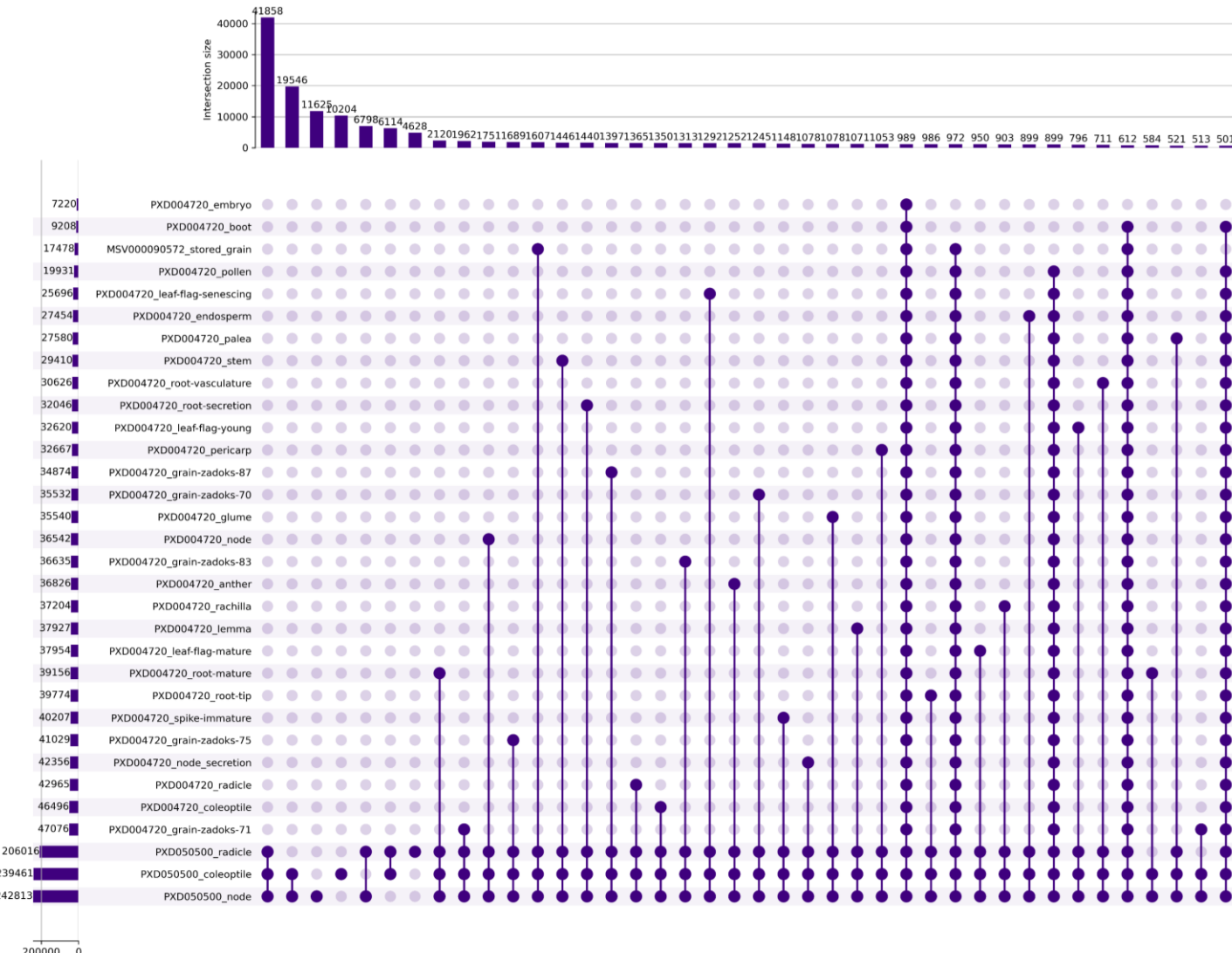
Supplementary Figure S2: Peptide support distributions across HC and LC wheat gene models. (A) Scatterplot showing the relationship between protein length and the number of projected unique peptides per protein isoform. (B) Boxplot comparing projected peptide support between HC and LC gene models. (C) Distribution of projected unique peptides per gene model showing broader and more abundant support for HC annotations.



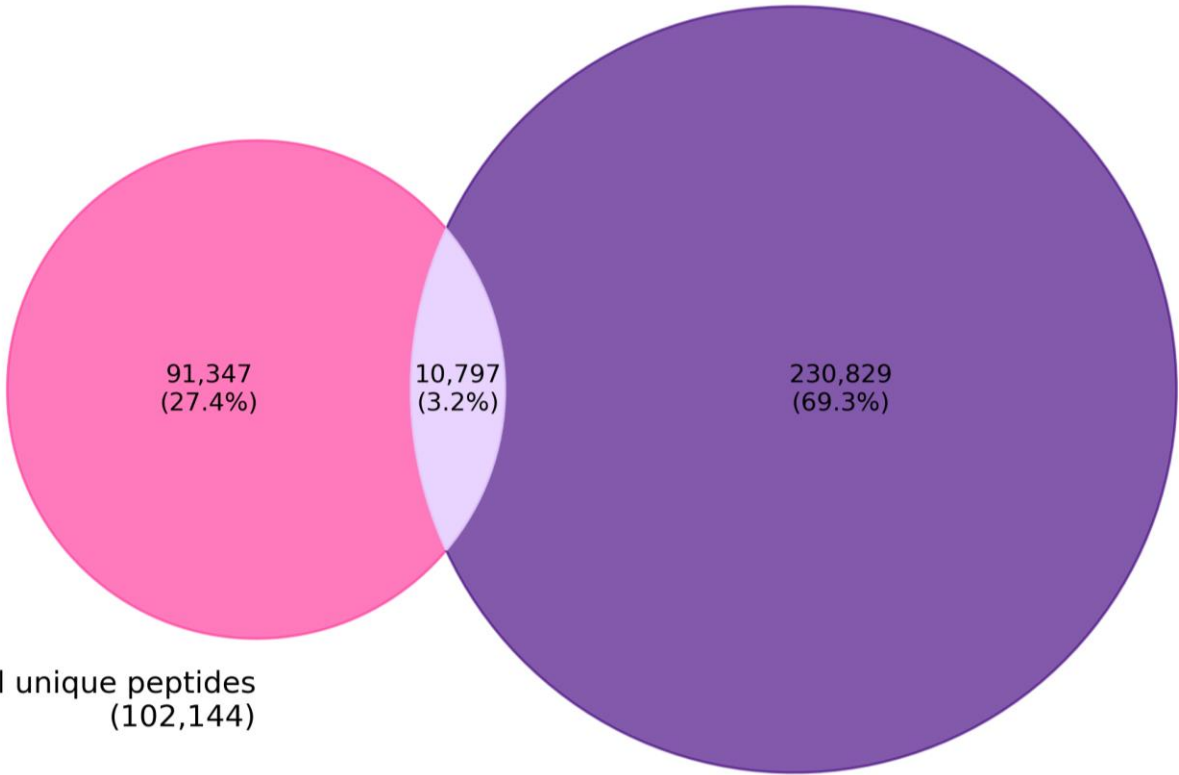
Supplementary Figure S3. Proteogenomic coverage across wheat tissues. Protein-level coverage showing the percentage of annotated HC and LC gene models supported by projected protein accessions.



Supplementary Figure S4: UpSet plot of projected protein isoform intersections across wheat tissues. Vertical bars indicate the number of shared projected protein isoforms among tissue combinations, while horizontal bars show the total number of projected protein isoforms identified per tissue. The analysis highlights both broadly shared and tissue-specific proteogenomic signatures across the wheat datasets.



Supplementary Figure S5: Comparison of unique peptide repertoires recovered by alignment-based and projection-based wheat proteogenomics workflows. The Venn diagram compares unique peptides identified using the exploratory tBLASTn alignment workflow of Vincent & Appels (2024) with the annotation-aware peptide projection workflow developed in this study. Only public datasets shared between both studies (MSV000090572 and PXD004720) were included for direct comparison. The limited overlap reflects substantial methodological differences between peptide-centric genomic alignment and GFF3-guided peptide projection approaches.



2024 total unique peptides
(102,144)

2026 total unique peptides
(241,626)